Complexity Theory

Course Project

Fine-Grained Complexity and Approximation Tradeoffs for Graph Diameter

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Abstract

Classical complexity theory classifies problems as polynomial-time solvable or NP-hard, but this coarse distinction cannot explain why some polynomial-time problems admit near-linear algorithms while others seem to require quadratic or cubic time. Fine-grained complexity addresses this gap by conjecturing precise lower bounds on natural problems, conditioned on standard hardness hypotheses, and by developing reductions that transmit these lower bounds between problems while preserving their running-time exponents. This report gives an introduction to fine-grained complexity, using the Diameter problem as a running example. We examine known results showing that computing the diameter exactly requires essentially quadratic time, even to distinguish diameter 2 from diameter 3, and that this hardness extends to a tight tradeoff between the time complexity and the approximation quality of diameter algorithms. Finally, we examine approximation algorithms that bypass these conditional lower bounds, illustrating how fine-grained complexity yields more precise statements about computational hardness than classical complexity theory.

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1 Introduction

The diameter of a graph $G = (V, E)$ is the maximum shortest-path distance between any pair of vertices. It is one of the most fundamental global parameters of a graph: in network applications, it measures the largest communication distance between two nodes, giving a basic indication of how quickly information can propagate through the network. While the diameter is trivial to define, computing it efficiently is not, and a significant gap remains between the best-known algorithms for the problem and its theoretical lower bounds.

A straightforward exact algorithm computes the diameter by running a breadth-first search (BFS) from every vertex. For an unweighted graph with n vertices and m edges, this takes $O(mn)$ time, which is quadratic in n once the graph is sparse ($m = O(n^{1+o(1)})$). Since Diameter asks only for the largest distance rather than all pairwise distances, it is natural to ask whether this quadratic barrier can be substantially improved.

Assuming the Strong Exponential Time Hypothesis (SETH), Roditty and Vassilevska Williams [RV13] proved that even distinguishing between diameter 2 and 3 in sparse unweighted graphs requires near-quadratic ($m^{2-o(1)}$) time. This effectively rules out not only a truly subquadratic exact algorithm, but also any truly subquadratic $(3/2 - \varepsilon)$-approximation, for every constant $\varepsilon > 0$.

On the algorithmic side, a single BFS trivially gives a factor-2 approximation, but beating this ratio in subquadratic time requires more structure. Aingworth, Chekuri, Indyk, and Motwani [**Ain+99**] were the first to do so, combining partial BFS computations with small dominating sets to obtain an almost-3/2 approximation.

The remainder of the report is organized as follows. Section 2 introduces the fine-grained complexity framework — SETH, fine-grained reductions, and the asymptotic notation used throughout — and fixes the definitions and approximation convention for Diameter. Section 3 examines the proof of the quadratic barrier above in detail, first via a reduction from Dominating Set and then via a direct reduction from CNF-SAT, the latter also giving the $(3/2-\varepsilon)$-approximation hardness stated above. Section 4 turns to the algorithmic side, presenting the algorithm of Aingworth et al. for breaking the factor-2 barrier. Section 5 discusses these results in the context of the broader literature, including the general tradeoff of Dalirrooyfard, Li, and Vassilevska Williams and more recent developments.

2 Preliminaries

2.1 Fine-Grained Complexity

Classical complexity theory separates problems into broad classes such as P and NP, but for problems already known to lie in P this separation is too coarse: it does not distinguish an $O(n)$-time algorithm from an $O(n^3)$-time one, although in practice this difference matters. *Fine-grained complexity* addresses this by asking whether the exponent c of a polynomial-time problem with running time $O(n^c)$ can be strictly improved to $O(n^{c-\delta})$ for some constant $\delta > 0$. Such improvements are usually not known to be impossible outright, so fine-grained complexity instead proves *conditional* lower bounds: it shows

that an improvement to the exponent for a given problem would imply an improvement for another problem whose exponent is conjectured, but not proven, to be optimal. This is analogous to the logic of NP-hardness, except that the object being transferred is not membership in P versus NP-hardness, but a specific polynomial exponent.

The Strong Exponential Time Hypothesis

As mentioned above, fine-grained complexity relies on the conjectured hardness of a small number of source problems. The source problem used throughout this report is the following.

Definition 2.1 (k-SAT)**.** For an integer $k \geq 1$, an instance of k-SAT is a Boolean formula φ in conjunctive normal form (a conjunction of clauses, each a disjunction of literals) over n variables, in which every clause contains at most k literals. The problem asks whether φ is satisfiable, i.e. whether some assignment of the n variables makes φ evaluate to true.

We use $\mathsf{CNF}-\mathsf{SAT}$ to denote the general satisfiability problem for Boolean formulas in CNF, without a bound on the number of literals per clause.

Exhaustive search solves k-SAT in $O(2^n \operatorname{poly}(n))$ time by trying all 2^n assignments. For each fixed k, algorithms with a smaller base of the exponent are known [Pat+05], although this base approaches 2 as $k \to \infty$.

Definition 2.2 (Strong Exponential Time Hypothesis)**.** The *Strong Exponential Time Hypothesis* (SETH) states that for every $\varepsilon > 0$ there exists an integer k such that k-SAT cannot be solved in $O(2^{(1-\varepsilon)n})$ time by any algorithm.

The reduction we examine later in this report encodes a k-SAT instance directly into a Diameter instance. We first describe, in general terms, the notion of reduction that lets such an encoding transfer a conditional lower bound from SETH to Diameter.

Fine-grained reductions

To transfer a conjectured lower bound from a source problem to a target problem, an ordinary polynomial-time reduction is not sufficient. This is because the complexity of the machine used to do the reduction does not matter as long as it is polynomial. Fine-grained reductions instead keep precise account of the size and number of oracle calls. This ensures that a hypothetical speed-up for the source problem translates into a speed-up for the target problem.

Definition 2.3 (Fine-grained reduction)**.** Let A and B be computational problems, and let $a(n_A)$ and $b(n_B)$ denote conjectured lower bounds on their running times as functions of their respective input sizes n_A and n_B. We say A *fine-grained reduces* to B, written $A \leq_{a,b} B$, if for every $\varepsilon > 0$ there exist a constant $\delta > 0$ and an algorithm that solves A on inputs of size n_A by making oracle calls to B on instances of sizes $n_{B,1}, \ldots, n_{B,q}$, such that the total time spent outside the oracle calls is $O((a(n_A))^{1-\delta})$ and

$$\sum_{i=1}^{q} (b(n_{B,i}))^{1-\varepsilon} \;=\; O\big((a(n_A))^{1-\delta}\big).$$

The bound on $\sum_i b(n_{B,i})^{1-\varepsilon}$ formalizes the following idea: if B could be solved in $O(b(n_{B,i})^{1-\varepsilon})$ time on every instance size $n_{B,i}$, then replacing each oracle call by a direct computation would solve A in $O(a(n_A)^{1-\delta})$ time, which is faster than the conjectured running time for A by the constant δ in the exponent. Consequently, if $A \leq_{a,b} B$ and no algorithm solves A in $O(a(n_A)^{1-\delta})$ time for any constant $\delta > 0$, then no algorithm solves B in $O(b(n_B)^{1-\varepsilon})$ time for any constant $\varepsilon > 0$ either.

In the remainder of this report, we apply this framework to a single running example. We examine a construction that transforms an instance of k-SAT into an instance of the Diameter problem; show that the construction produces a specific gap in the diameter value depending on the satisfiability of the formula; and conclude that a sufficiently fast algorithm for Diameter would violate SETH.

Notation for fine-grained bounds

Let n denote the size of the input, and let $c > 0$ be a constant. We use standard asymptotic notation to describe running times, focusing on distinctions up to subpolynomial or polylogarithmic factors.

- The expression $n^{c\pm o(1)}$ denotes a quantity of the form $n^{c\pm\varepsilon(n)}$ for some strictly positive function $\varepsilon(n) \to 0$ as $n \to \infty$. This means the polynomial exponent asymptotically approaches c. Thus, for any constant $\delta > 0$ and all sufficiently large n,

  $$n^{c-\delta} \leq n^{c-o(1)} \leq n^{c+\delta}.$$

  This notation is frequently used to safely absorb polylogarithmic factors into the exponent. To see why this works, observe that for any constant $k > 0$, a polylogarithmic factor $\log^k n$ can be rewritten as a base-n exponential:

  $$\log^k n = n^{\frac{k\log(\log n)}{\log n}}.$$

  Letting $\varepsilon(n) = \frac{k\log(\log n)}{\log n}$, we can see that $\varepsilon(n) \to 0$ as $n \to \infty$, meaning $\varepsilon(n) \in o(1)$. Consequently, polylogarithmic factors are strictly subsumed by the $o(1)$ exponent:

  $$n^c \log^{O(1)} n \subseteq n^{c+o(1)}.$$

  In the context of conditional lower bounds, an $n^{2-o(1)}$ lower bound rules out *any* algorithm running in $O(n^{2-\delta})$ time for every constant $\delta > 0$.

- For upper bounds, we often write $g(n) = \widetilde{O}(f(n))$ to suppress polylogarithmic factors, meaning:

  $$g(n) \in O\Big(f(n)\log^{O(1)} n\Big).$$

  Unlike $n^{o(1)}$, which can represent *any* subpolynomial factor (such as $2^{\sqrt{\log n}}$ or $n^{1/\log\log n}$), the $\widetilde{O}$ notation conventionally hides *only* polylogarithmic terms.

2.2 The Diameter Problem

We now introduce the formal graph notation and approximation conventions used throughout the report. All graphs discussed in this project are assumed to be unweighted. Unless

stated otherwise, let $G = (V, E)$ be an unweighted, undirected graph with $n = |V|$ vertices and $m = |E|$ edges. In Section 4, we will extend these definitions to the directed setting.

Definition 2.4 (Distance)**.** For vertices $u, v \in V$, the *distance* $\mathrm{dist}(u, v)$ is defined as the minimum number of edges on a path from u to v. If no such path exists, we set $\mathrm{dist}(u, v) = \infty$.

Definition 2.5 (Eccentricity)**.** For a vertex $v \in V$, its *eccentricity* is defined as the maximum distance to any other vertex in the graph, $\mathrm{ecc}(v) = \max_{u \in V} \mathrm{dist}(v, u)$.

Definition 2.6 (Diameter)**.** The *diameter* of a graph G is the maximum distance between any pair of vertices, $\mathrm{diam}(G) = \max_{u,v \in V} \mathrm{dist}(u, v)$. A pair of vertices u, v satisfying $\mathrm{dist}(u, v) = \mathrm{diam}(G)$ is called a *diametral pair*.

Definition 2.7 (The Diameter Problem)**.** In the Diameter problem, we are given a graph $G = (V, E)$, and the task is to compute $\mathrm{diam}(G)$.

The straightforward exact algorithm for Diameter executes a Breadth-First Search (BFS) from every vertex, requiring $O(mn)$ time. For sparse graphs ($m = O(n)$), the total size of the input is proportional to n, yielding an $O(n^2)$ running time that is quadratic in the input size.

The subsequent sections study whether this quadratic barrier is fundamental, and to what extent it can be bypassed by settling for an approximate answer.

Definition 2.8 (Diameter Approximation)**.** For a factor $\alpha \geq 1$, an *α-approximation* for Diameter is an algorithm that returns an estimate $\widehat{D}$ such that $\mathrm{diam}(G)/\alpha \leq \widehat{D} \leq \mathrm{diam}(G)$.

To establish conditional lower bounds for approximation, we will examine fine-grained reductions that construct graphs whose diameters differ by a multiplicative gap based on an underlying hard problem. If a reduction outputs a graph with diameter at most D_0 in the "yes" case, and a graph with diameter at least $D_1 > D_0$ in the "no" case, then achieving any approximation factor strictly smaller than D_1/D_0 is sufficient to distinguish the two cases. This observation turns distance gaps into conditional hardness results for approximation.

3 The Quadratic Barrier for Diameter in Sparse Graphs

Roditty and Vassilevska Williams [RV13] showed that, assuming SETH, there is no truly subquadratic algorithm for Diameter on *sparse* ($m = O(n^{1+o(1)})$) undirected unweighted graphs. While the standard $O(mn)$ BFS algorithm is trivially subquadratic relative to the input size on dense graphs (running in $O(m^{1.5})$ time), breaking the quadratic barrier remains conditionally impossible for sparse graphs. In fact, even distinguishing graphs of diameter 2 from graphs of diameter 3 requires $m^{2-o(1)}$ time.

We first examine a reduction from a fixed-size variant of Dominating Set, which illustrates the main construction and proves that no $O(m^{2-\varepsilon})$ algorithm exists. However, because the hard graph instances generated by this first reduction are not sparse, our

work is not done yet. To properly establish this barrier for sparse graphs, we then examine the direct reduction from CNF-SAT, which establishes the same 3/2 approximation barrier directly under SETHusing sufficiently sparse graphs.

3.1 A Reduction from Dominating Set

Definition 3.1. Let $G = (V, E)$ be a graph. A set $S \subseteq V$ *dominates* a vertex $v \in V$ if either $v \in S$ or v has a neighbor in S. The set S is a *dominating set* if it dominates every vertex of G. For an integer $\ell \geq 1$, ℓ-Dominating Set asks whether a given graph has a dominating set of size ℓ.

For every fixed integer ℓ, brute force solves ℓ-Dominating Set in $O(n^{\ell+O(1)})$ time, where n is the number of vertices. Moreover, under SETH, for every $\varepsilon > 0$ and every fixed ℓ, there is no $O(n^{\ell-\varepsilon})$-time algorithm for this problem [PW10].

Fix a constant $k \geq 1$ and set $\ell = 2k$. A dominating set of size ℓ splits into two subsets of size k, so it suffices to detect when the union of two k-subsets of V dominates G.

Construction

Let $G = (V, E)$ be an n-vertex instance of $2k$-Dominating Set, and let $\mathcal{S}_k$ denote the family of all k-element subsets of V. We construct a graph G' as follows.

- For every $S \in \mathcal{S}_k$, add a vertex to G'. These vertices are pairwise non-adjacent.
- Add a copy V' of V, and make V' a clique.
- For every $S \in \mathcal{S}_k$ and $v \in V'$, add the edge $\{S, v\}$ if and only if S does not dominate v in G.

Whether some $S \in \mathcal{S}_k$ already dominates G can be checked in $O(n^{k+1})$ time: there are $O(n^k)$ choices for S, and checking whether a given S dominates all n vertices takes $O(n)$ time. If so, any superset of S of size $2k$ is dominating, and we answer directly. Hence, assume that no $S \in \mathcal{S}_k$ dominates G, so every subset-vertex has at least one neighbor in V'.

Distance Gap

Consider two subset-vertices $S, T \in \mathcal{S}_k$ with $S \neq T$.

Suppose $S \cup T$ does not dominate G. Then some $u \in V$ is dominated by neither S nor T, so both are adjacent to $u \in V'$, giving $\mathrm{dist}_{G'}(S, T) = 2$ (recall S, T are not adjacent).

Suppose $S \cup T$ dominates G. Since neither S nor T dominates G alone, there is $u \in V$ not dominated by S and $w \in V$ not dominated by T. Then S–u–w–T is a path of length 3, using that V' is a clique. No shorter path exists: a common neighbor of S, T would be a vertex dominated by neither, contradicting that $S \cup T$ dominates G. Hence $\mathrm{dist}_{G'}(S, T) = 3$.

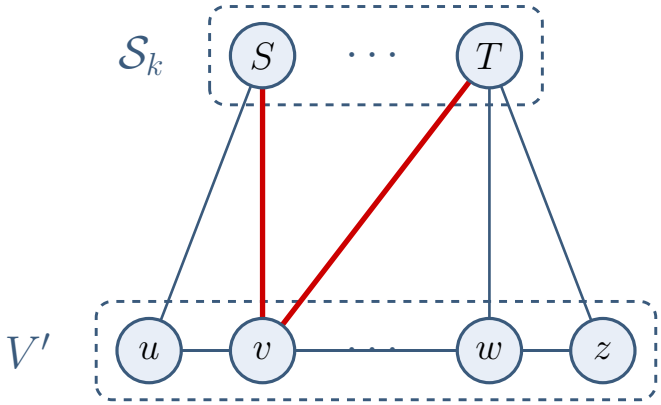


Figure 1: The construction from Dominating Set. The vertices in $\mathcal{S}_k$ have no edges between them, while V' forms a clique. For $S, T \in \mathcal{S}_k$ their common neighbor $v \in V'$ represents a vertex of G that is dominated by neither S nor T, giving $\mathrm{dist}_{G'}(S, T) = 2$.

Every other pair of vertices in G' is at distance at most 2, since V' is a clique and every subset-vertex has a neighbor in V'. Therefore,

$$\mathrm{diam}(G') = \begin{cases} 3, & \text{if } G \text{ has a dominating set of size } 2k, \\ 2, & \text{otherwise.} \end{cases}$$

Size Analysis

The graph G' has $\binom{n}{k} + n = O(n^k)$ vertices. Its edges consist of the $O(n^2)$ clique edges on V', together with at most one edge between each subset-vertex and each vertex of V'. Thus,

$$|E(G')| = O(n^{k+1}).$$

Observe that this constructed graph is not sparse, as the number of edges $O(n^{k+1})$ is strictly larger than the number of vertices $O(n^k)$ by a factor of n, meaning $m \notin O(|V(G')|)$. Moreover, G' can be constructed in $O(n^{k+1})$ time by considering each of the $O(n^k)$ subset-vertices and determining its adjacency to each of the n vertices of V'.

This gives a fine-grained reduction

$$\ell\text{-Dominating Set} \leq_{n^\ell, m^2} \text{Diameter}_{\text{2-vs-3}},$$

where $\ell = 2k$, n is the number of vertices in the ℓ-Dominating Set instance, and m is the number of edges in the constructed graph.

Suppose that $\text{Diameter}_{\text{2-vs-3}}$ admits an $O(m^{2-\varepsilon})$-time algorithm for some $\varepsilon > 0$. Since $m = O(n^{k+1})$, applying this algorithm to G' takes

$$O\big(n^{(k+1)(2-\varepsilon)}\big) = O\Big(n^{2k-\varepsilon'}\Big), \qquad \varepsilon' = \varepsilon(k+1) - 2.$$

Choosing $k \geq \lceil 2/\varepsilon \rceil$ gives $\varepsilon' > 0$, and hence yields an $O(n^{\ell-\varepsilon'})$-time algorithm for ℓ-Dominating Set, contradicting its conjectured n^ℓ lower bound.

3.2 A Direct Reduction from CNF-SAT

Theorem 3.2 (Roditty–Vassilevska Williams [RV13]). *Assuming* SETH, *for every* $\varepsilon > 0$ *and* $\delta > 0$, *there is no* $O(M^{2-\varepsilon})$*-time* $(3/2-\delta)$*-approximation algorithm for* DIAMETER *on undirected unweighted graphs with* M *edges.*

Proof. **Construction**

Let φ be a CNF-SATformula with n variables and m_φ clauses. Partition the variables into X_1 and X_2, with $n_1 = \lceil n/2 \rceil$ and $n_2 = n - n_1$ variables, and let A_i be the set of all 2^{n_i} truth assignments to X_i, for $i = 1, 2$.

We construct a graph G' as follows.

- For each assignment $\alpha \in A_1$, add a vertex v_α. Denote these vertices by $\alpha_1, \ldots, \alpha_{2^{n_1}}$. Similarly, for each assignment $\beta \in A_2$, add a vertex v_β, denoted by $\beta_1, \ldots, \beta_{2^{n_2}}$.
- Add two vertices t_1, t_2. Connect t_1 to every α_i, and t_2 to every β_i.
- For every clause c_i, $1 \le i \le m_varphi$, add a vertex c_i. Make t_1, t_2 and all clause vertices $c_1, \ldots, c_{m_\varphi}$ pairwise adjacent, forming a clique.
- Connect an assignment vertex to a clause vertex exactly when the corresponding assignment does not satisfy that clause.

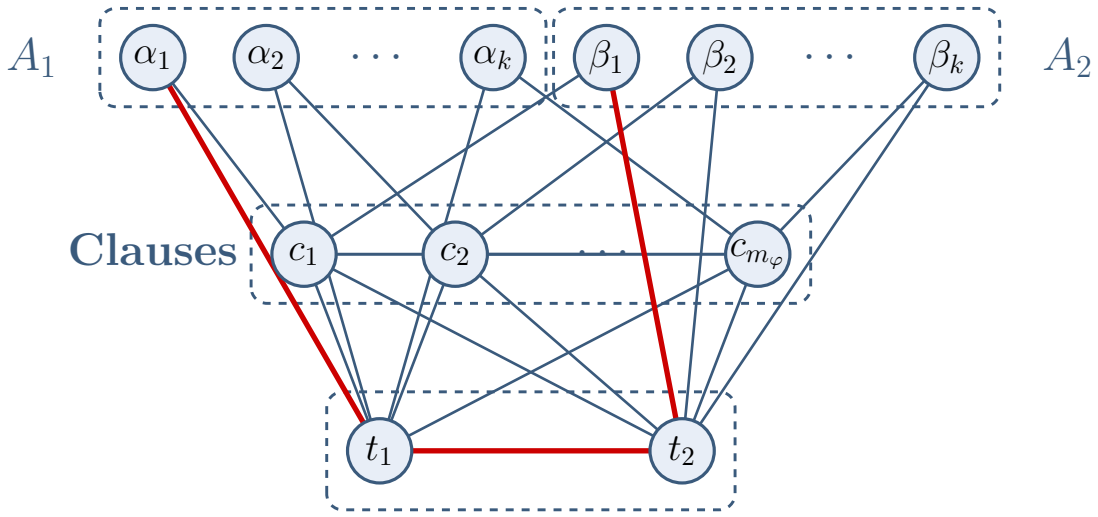


Figure 2: Illustration of the construction for Theorem 3.2. The assignment vertices $\alpha_1, \ldots, \alpha_k \in A_1$ and $\beta_1, \ldots, \beta_k \in A_2$ are connected to the clauses they falsify, while each t_i connects to all vertices in its respective set A_i. Furthermore, t_1, t_2, and all clause vertices together form a clique. The highlighted edges demonstrate a sample path of length 3 between α_1 and β_1.

Correctness

Fix $\alpha \in A_1$ and $\beta \in A_2$. We show that $\mathrm{dist}_{G'}(v_\alpha, v_\beta) = 3$ if $\alpha \cup \beta$ satisfies φ, and $\mathrm{dist}_{G'}(v_\alpha, v_\beta) = 2$ otherwise. We also show that every other pair of vertices is at distance

at most 2.

If $\alpha \cup \beta$ satisfies φ, every clause is satisfied by α or β. Hence, no clause vertex is adjacent to both v_α and v_β. Since the two assignment vertices are also non-adjacent, $\mathrm{dist}_{G'}(v_\alpha, v_\beta) \geq 3$. The path

$$v_\alpha - t_1 - t_2 - v_\beta$$

gives equality.

If $\alpha \cup \beta$ does not satisfy φ, some clause c is falsified by both assignments. Thus, v_c is adjacent to both v_α and v_β, giving $\mathrm{dist}_{G'}(v_\alpha, v_\beta) = 2$.

For all remaining pairs, t_1, t_2 and the clause vertices form a clique; two assignment vertices in the same A_i are both adjacent to t_i; and an assignment vertex v_γ, with $\gamma \in A_i$, is either adjacent to a clause vertex v_c, or reaches it via $v_\gamma - t_i - v_c$. Finally, v_γ reaches t_{3-i} through t_i. Thus, every such pair is at distance at most 2, and

$$\mathrm{diam}(G') = \begin{cases} 3, & \text{if } \varphi \text{ is satisfiable,} \\ 2, & \text{otherwise.} \end{cases}$$

Size and Running Time

The graph G' has $O(m_\varphi + 2^{n/2})$ vertices. Its edges consist of $O(m_\varphi^2)$ clique edges together with at most one edge between each assignment vertex and each clause vertex. Hence, for $m_\varphi = \mathrm{poly}(n)$,

$$|E(G')| = O(m_\varphi \cdot 2^{n/2}),$$

where m_φ is the number of clauses in the CNF-SATinstance. Since the number of vertices is $|V(G')| = O(m_\varphi + 2^{n/2})$ and $m_\varphi = \mathrm{poly}(n)$, the number of edges is bounded by $|V(G')| \,\mathrm{poly}(\log |V(G')|)$; this means $|E(G')| = |V(G')|^{1+o(1)}$, making this constructed graph sufficiently sparse.

Let $N = |V(G')| + |E(G')|$ denote the total input size of the constructed graph. Because the graph is connected and sparse, the input size is dominated by the number of edges, meaning $N = \Theta(|E(G')|) = O(m_\varphi \cdot 2^{n/2})$.

This gives a fine-grained reduction directly in terms of the input size:

$$\mathsf{CNF} - \mathsf{SAT} \leq_{2^n, N^2} \textsc{Diameter}_{\text{2-vs-3}},$$

where n is the number of variables in the CNF-SATinstance and N is the total size of the constructed Diameter instance.

Suppose that $\textsc{Diameter}_{\text{2-vs-3}}$ admits an $O(N^{2-\varepsilon})$-time algorithm (that is, an algorithm truly subquadratic relative to its input size) for some $\varepsilon > 0$. Since $N = O(m_\varphi \cdot 2^{n/2})$, applying this algorithm to G' takes

$$O\big(N^{2-\varepsilon}\big) = O\big((m_\varphi \cdot 2^{n/2})^{2-\varepsilon}\big) = O\big(2^{n(1-\varepsilon/2)}\,\mathrm{poly}(n)\big)\,.$$

This would give a truly faster-than-2^n algorithm for CNF-SAT, contradicting SETH. □

4 Approximation Algorithms for Diameter

The previous section showed that the quadratic barrier for DIAMETER extends beyond *exact* computation: under SETH, the reduction in Theorem 3.2 rules out any truly subquadratic $(3/2-\delta)$-approximation for any $\delta > 0$. Thus, any subquadratic algorithm is conditionally forced to have an approximation ratio of at least $3/2$. While a single BFS trivially provides a 2-approximation, this leaves a notable gap between the conditional lower bound of $3/2$ and the achievable upper bound of 2. In this section, we study algorithms that narrow this gap, tracing the development of increasingly accurate DIAMETER approximations that approach the $3/2$ barrier.

Proposition 4.1 (Single-source factor-2 approximation)**.** *Let $G = (V, E)$ be an unweighted undirected graph of finite diameter, and let $s \in V$ be arbitrary. Then* $\operatorname{diam}(G)/2 \leq \operatorname{ecc}(s) \leq \operatorname{diam}(G)$*. Consequently, one BFS gives a factor-2 approximation of the diameter.*

Proof. Let $D = \operatorname{diam}(G)$, and let u, v be a diametral pair. By the triangle inequality, $D \leq \operatorname{dist}(u, s) + \operatorname{dist}(s, v)$. Since the graph is undirected, both terms are bounded by $\operatorname{ecc}(s)$, and hence $D \leq 2\operatorname{ecc}(s)$. The upper bound $\operatorname{ecc}(s) \leq D$ follows directly from the definition of diameter. □

The first major improvement over this basic factor-2 approximation was given by Aingworth, Chekuri, Indyk, and Motwani [Ain+99]. Their algorithm achieves an almost-$3/2$ approximation by avoiding a full BFS computation from every vertex. Instead, it constructs a small collection of carefully chosen roots and guarantees that, for every diametral pair, one of these roots is sufficiently close to one of its endpoints. In this section, we discuss their algorithm and its analysis.

4.1 Breaking the Factor-2 Barrier

Throughout this subsection, let $G = (V, E)$ be an unweighted directed graph with n vertices and m edges, and assume that G has finite diameter. Let $\ell = \operatorname{diam}(G)$.

For a vertex v, let $\operatorname{ecc}^+(v) = \max_{u \in V} \operatorname{dist}(v, u)$ and $\operatorname{ecc}^-(v) = \max_{u \in V} \operatorname{dist}(u, v)$ denote its out- and in-eccentricities, respectively. For an integer $r \geq 0$, define $N_r^+(v) = \{u \in V \setminus \{v\} : \operatorname{dist}(v, u) \leq r\}$ and $N_r^-(v) = \{u \in V \setminus \{v\} : \operatorname{dist}(u, v) \leq r\}$.

Fix a diametral ordered pair a, b, so that $\operatorname{dist}(a, b) = \ell$. Suppose for the moment that ℓ is divisible by 3, and write $r = \ell/3$.

If $x \in N_r^+(a)$, then the triangle inequality gives $\operatorname{dist}(x, b) \geq \operatorname{dist}(a, b) - \operatorname{dist}(a, x) \geq \ell - r = 2\ell/3$. Consequently, $\operatorname{ecc}^+(x) \geq 2\ell/3$. Symmetrically, if $y \in N_r^-(b)$, then $\operatorname{dist}(a, y) \geq \ell - r = 2\ell/3$, and hence $\operatorname{ecc}^-(y) \geq 2\ell/3$.

Thus, to obtain a $2/3$-estimate of the diameter, it is enough to perform a forward shortest-path search from some vertex in $N_r^+(a)$, or a reverse shortest-path search from some vertex in $N_r^-(b)$. The difficulty is that the diametral pair a, b is not known in advance.

Definition 4.2 (Partial BFS)**.** For $1 \leq k < n$, a k-partial-BFS rooted at v is obtained by running a forward BFS from v until exactly k vertices other than v have been discovered.

Ties between vertices at the same distance are broken arbitrarily.

Let $\mathrm{PBFS}_k^+(v)$ denote these k vertices, and let $\mathrm{pb}_k^+(v)$ denote the depth of the resulting partial-BFS tree. We also write $\overline{\mathrm{PBFS}}_k^+(v) = \mathrm{PBFS}_k^+(v) \cup \{v\}$.

The definition immediately gives two useful properties. For every integer r, if $|N_r^+(v)| \geq k$, then $\mathrm{PBFS}_k^+(v) \subseteq N_r^+(v)$; if $|N_r^+(v)| < k$, then $N_r^+(v) \subseteq \mathrm{PBFS}_k^+(v)$.

Lemma 4.3. *A k-partial-BFS can be computed in $O(k^2)$ time.*

Proof. Before the search terminates, at most $k+1$ vertices have been discovered. For each expanded vertex, at most $O(k)$ edges can lead to already discovered vertices, while at most k edges in total discover new vertices before the search stops. Hence only $O(k^2)$ edges are examined. □

Directional domination. We use the convention of Aingworth et al. [Ain+99]. For an undirected graph, a set $D \subseteq V$ is dominating if every vertex outside D has a neighbor in D. In a directed graph, a set $D^+ \subseteq V$ is *out-dominating* if every $v \in V \setminus D^+$ has an outgoing edge to a vertex of D^+; that is, there exists $u \in D^+$ such that $(v,u) \in E$. Symmetrically, $D^- \subseteq V$ is *in-dominating* if every $v \in V \setminus D^-$ has an incoming edge from a vertex of D^-; that is, there exists $u \in D^-$ such that $(u,v) \in E$.

The following theorem provides the small dominating sets used by the algorithm.

Theorem 4.4 (Aingworth et al. [Ain+99])**.** *Let $G = (V,E)$ be a graph, and let*

$$L(V) = \{v \in V : d_v < s\}, \qquad H(V) = V \setminus L(V).$$

There exists a dominating set of $H(V)$ of size $O\left(\frac{n \log n}{s}\right)$, and such a set can be computed in $O(m + ns)$ time.

Algorithm 1 Approx-Diameter

1: Compute an s-partial forward BFS from every $v \in V$
2: Let $w \in \arg\max_{v \in V} \mathrm{pb}_s^+(v)$
3: Compute a forward BFS from w, and a reverse BFS from every vertex in $\overline{\mathrm{PBFS}}_s^+(w)$
4: Construct $\widehat{G}$ from G by adding (v,u) for every $v \in V$ and $u \in \mathrm{PBFS}_s^+(v)$
5: Compute an out-dominating set D^+ of $\widehat{G}$
6: Compute a forward BFS from every vertex in D^+
7: **return** the maximum depth among the BFS trees computed in Steps 3 and 6

We now analyze why the algorithm works. For clarity, we continue to assume that $\ell \geq 3$ and that $r = \ell/3$ is an integer.

Lemma 4.5 (Large forward neighborhoods)**.** *The set D^+ computed in Step 5 satisfies $|D^+| = O(n \log n/s)$. Moreover, if $|N_r^+(v)| \geq s$ for every $v \in V$, then for every $v \in V$, $D^+ \cap (N_r^+(v) \cup \{v\}) \neq \varnothing$.*

Proof. By construction of $\widehat{G}$, every vertex has out-degree at least s. Therefore, the directed version of Theorem 4.4 gives $|D^+| = O(n \log n/s)$.

Now fix $v \in V$. If $v \in D^+$, the second claim is immediate. Otherwise, since D^+ out-dominates $\widehat{G}$, there exists $u \in D^+$ such that $(v,u) \in E(\widehat{G})$.

If $(v,u) \in E(G)$, then $\mathrm{dist}(v,u) = 1 \le r$, and hence $u \in N_r^+(v)$. Otherwise the edge was added in Step 4, so $u \in \mathrm{PBFS}_s^+(v)$. Since $|N_r^+(v)| \ge s$, the s closest vertices to v all lie within distance r, and therefore $\mathrm{PBFS}_s^+(v) \subseteq N_r^+(v)$. In either case, $u \in D^+ \cap N_r^+(v)$. □

The complementary situation is exactly why the algorithm chooses the vertex w whose partial BFS has maximum depth.

Lemma 4.6 (A small forward neighborhood). *Let $S = \{v \in V : |N_r^+(v)| < s\}$. If $S \neq \varnothing$, then the vertex w selected in Step 2 belongs to S.*

Moreover, if $\mathrm{ecc}^+(w) < 2r$, then for every $v \in V$,

$$\overline{\mathrm{PBFS}}_s^+(w) \cap \left(N_r^-(v) \cup \{v\}\right) \neq \varnothing.$$

Proof. If $u \in S$, fewer than s vertices lie within distance r of u. Hence the s-partial BFS from u must go beyond distance r, and so $\mathrm{pb}_s^+(u) > r$.

On the other hand, if $v \notin S$, at least s vertices are within distance r, and therefore $\mathrm{pb}_s^+(v) \le r$. Thus every vertex maximizing the partial-BFS depth belongs to S, in particular $w \in S$.

Since $w \in S$, every vertex of $N_r^+(w)$ is discovered before the s-partial BFS terminates. Hence $N_r^+(w) \subseteq \mathrm{PBFS}_s^+(w)$.

Now suppose that $\mathrm{ecc}^+(w) < 2r$, and fix $v \in V$. Then $\mathrm{dist}(w,v) < 2r$. Consider a shortest path from w to v. If $\mathrm{dist}(w,v) \le r$, choose $x = v$. Otherwise, choose the vertex x on this path at distance exactly r from w. In the latter case, $\mathrm{dist}(x,v) < r$.

Therefore $x \in \overline{\mathrm{PBFS}}_s^+(w)$ and $x \in N_r^-(v) \cup \{v\}$, proving the claim. □

The two preceding lemmas correspond to the two possible structures of the graph: either every radius-r forward neighborhood is large, in which case the dominating set hits the neighborhood of the first diametral endpoint, or some such neighborhood is small, in which case the maximum-depth partial BFS identifies a useful vertex w.

Theorem 4.7 (Aingworth et al. [Ain+99]). *Let $G = (V,E)$ be an unweighted directed graph with n vertices, m edges, and diameter ℓ. If ℓ is divisible by 3, Algorithm 1 returns an estimate $\widehat{\ell}$ satisfying $\frac{2\ell}{3} \le \widehat{\ell} \le \ell$.*

For a parameter s, its running time is $O\left(ns^2 + ms + \frac{mn\log n}{s} + n^2\right)$. In particular, choosing $s = \lceil\sqrt{n\log n}\rceil$ gives $O(m\sqrt{n\log n} + n^2\log n)$ time.

Proof. Every value considered by the algorithm is the depth of a forward or reverse shortest-path tree in G. Hence $\widehat{\ell} \le \ell$.

For the lower bound, fix a diametral ordered pair a, b, so that $\mathrm{dist}(a,b) = \ell$, and again write $r = \ell/3$.

Case 1: every radius-r forward neighborhood is large. Suppose that $|N_r^+(v)| \geq s$ for every $v \in V$. By Lemma 4.5, there exists $x \in D^+ \cap (N_r^+(a) \cup \{a\})$.

If $x = a$, then $\mathrm{ecc}^+(x) \geq \mathrm{dist}(a,b) = \ell$. Otherwise, $\mathrm{dist}(a,x) \leq r$, and therefore $\mathrm{dist}(x,b) \geq \mathrm{dist}(a,b) - \mathrm{dist}(a,x) \geq \ell - r = 2\ell/3$. Thus in either case $\mathrm{ecc}^+(x) \geq 2\ell/3$.

Step 6 performs a forward BFS from every vertex in D^+, including x, so the returned estimate is at least $2\ell/3$.

Case 2: some radius-r forward neighborhood is small. Suppose that $|N_r^+(v)| < s$ for some $v \in V$. Then $S \neq \varnothing$, and by Lemma 4.6, the vertex w chosen in Step 2 belongs to S.

If $\mathrm{ecc}^+(w) \geq 2\ell/3$, the forward BFS from w in Step 3 already gives the desired estimate. We may therefore assume that $\mathrm{ecc}^+(w) < 2\ell/3 = 2r$.

Applying Lemma 4.6 with $v = b$, there exists $x \in \overline{\mathrm{PBFS}}_s^+(w)$ such that $x \in N_r^-(b) \cup \{b\}$.

If $x = b$, then $\mathrm{ecc}^-(x) \geq \mathrm{dist}(a,b) = \ell$. Otherwise, $\mathrm{dist}(x,b) \leq r$, and therefore $\mathrm{dist}(a,x) \geq \mathrm{dist}(a,b) - \mathrm{dist}(x,b) \geq \ell - r = 2\ell/3$. Hence in either case $\mathrm{ecc}^-(x) \geq 2\ell/3$.

Step 3 performs a reverse BFS from x, so the algorithm again finds a value of at least $2\ell/3$. This proves the approximation guarantee.

It remains to bound the running time. By Lemma 4.3, the n partial BFS searches in Step 1 take $O(ns^2)$ time. Selecting w in Step 2 takes $O(n)$ time. Step 3 performs $O(s)$ full BFS searches and therefore costs $O(ms)$.

The auxiliary graph in Step 4 contains at most ns added edges. Using the auxiliary dominating-set procedure, Step 5 takes $O(n^2 + ns)$ time. By Lemma 4.5, $|D^+| = O(n \log n/s)$, so the forward BFS searches in Step 6 cost $O(mn \log n/s)$.

Combining these bounds gives $O(ns^2 + ms + mn \log n/s + n^2)$. Finally, for $s = \lceil \sqrt{n \log n} \rceil$, both BFS terms become $O(m\sqrt{n \log n})$, while $ns^2 = O(n^2 \log n)$, giving the claimed running time. □

For diameters not divisible by 3, the same argument requires the appropriate integer rounding of the radius. In the unweighted case, the standard guarantee can be stated as $\lfloor 2\ell/3 \rfloor \leq \widehat{\ell} \leq \ell$.

5 Discussion and Conclusion

Here, we contextualize our main results within the broader field and conclude by highlighting the questions they leave open.

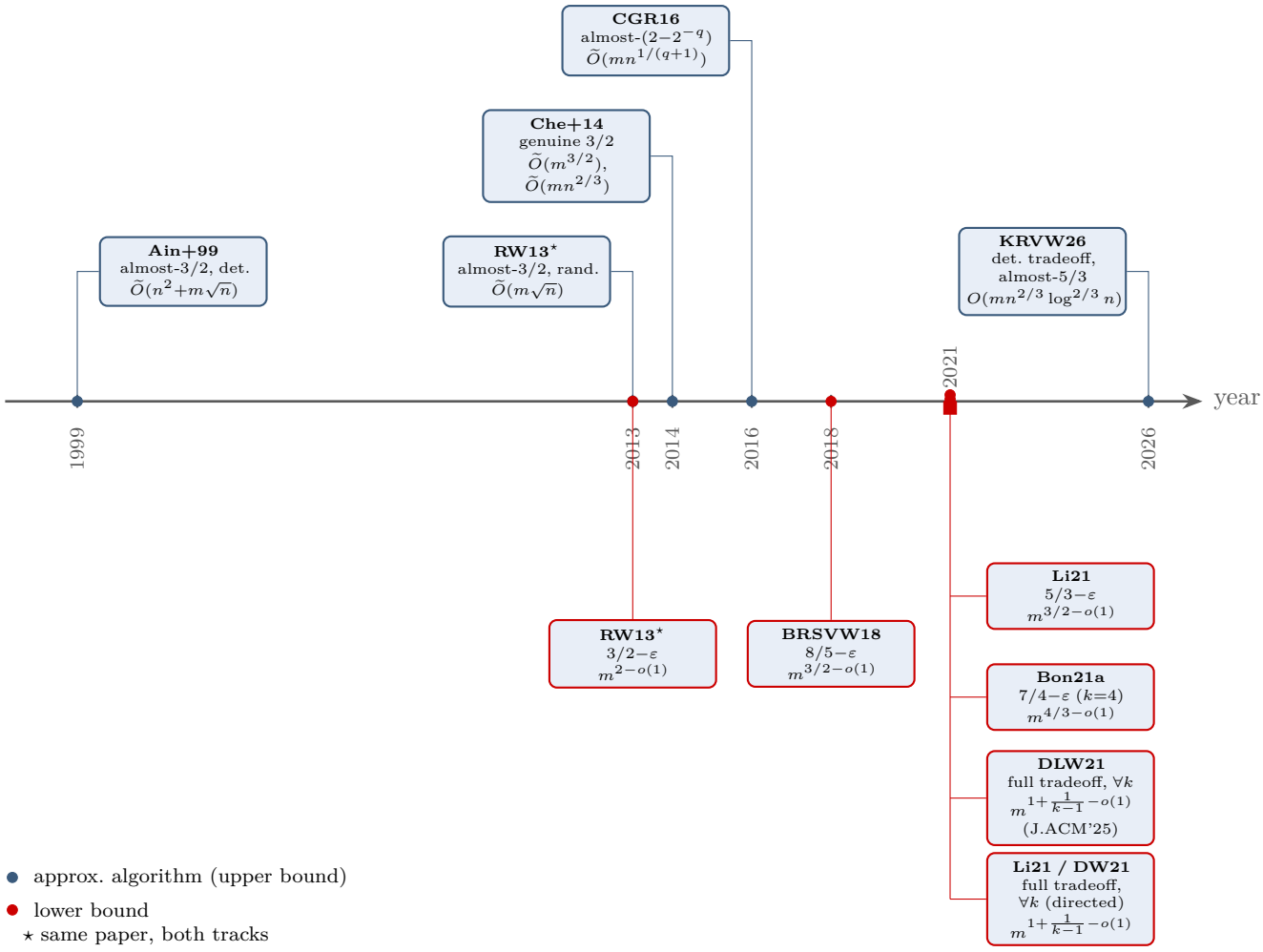


Figure 3: Timeline of the main results mentioned in the text. The upper half traces algorithms breaking the factor-2 barrier, while the lower half tracks the SETH-based conditional lower bounds that ultimately match them via full tradeoff curves.

Approximation algorithms. The first substantial improvement over the elementary factor-2 approximation was obtained by Aingworth, Chekuri, Indyk, and Motwani [**Ain+99**]. They gave a deterministic algorithm running in $\widetilde{O}(n^2+m\sqrt{n})$ time that achieves an almost-3/2 approximation, where the guarantee contains a small additive error in the weighted setting. Roditty and Vassilevska Williams [RV13] later obtained the same guarantee in expected $\widetilde{O}(m\sqrt{n})$ time using a randomized sampling approach.

Chechik et al. [**Che+14**] removed the additive error and obtained genuine 3/2-approximation algorithms with running times $\widetilde{O}(m^{3/2})$ and $\widetilde{O}(mn^{2/3})$. Thus, the factor-3/2 point can be achieved in truly subquadratic time on sufficiently sparse graphs, although the running time is higher than that of the almost-3/2 algorithms.

Cairo, Grossi, and Rizzi [CGR16] extended these ideas to an entire approximation–time tradeoff for undirected graphs. For every integer $q \geq 0$, their randomized scheme obtains an almost-$(2-2^{-q})$ approximation in expected $\widetilde{O}(mn^{1/(q+1)})$ time. As q increases, the running time approaches linear while the approximation factor approaches 2. This has remained the main general upper-bound tradeoff for undirected Diameter.

Very recently, Kirkpatrick, Roditty, Qi, and Vassilevska Williams [Kir+26] connected Diameter approximation to approximate distance oracles. They derandomized the Cairo–Grossi–Rizzi scheme without worsening its asymptotic running time, giving the first deterministic version of this tradeoff. They also gave a simpler deterministic genuine 3/2-approximation running in $O(mn^{2/3}\log^{2/3} n)$ time and obtained a new deterministic almost-5/3 approximation point. These results show that distance-oracle techniques provide another useful approach to Diameter approximation, although the known upper bounds still do not match the conditional lower-bound curve.

Conditional lower bounds. The hardness picture developed in parallel. Roditty and Vassilevska Williams [RV13] established the first central fine-grained lower bound: under SETH, a $(3/2-\varepsilon)$-approximation for sparse undirected unweighted Diameter requires $m^{2-o(1)}$ time. This is the 2-versus-3 gap studied in section 4.

Backurs, Roditty, Segal, Vassilevska Williams, and Wein [Bac+18] subsequently strengthened the hardness at smaller running-time exponents. For undirected unweighted graphs, they ruled out an $(8/5-\varepsilon)$-approximation in $m^{3/2-o(1)}$ time; their weighted construction obtained the stronger 5/3 threshold. Li [Li21] later improved the unweighted lower bound to $5/3-\varepsilon$ with the same $m^{3/2-o(1)}$ running-time barrier.

The next point on the undirected curve was obtained by Bonnet [Bon21], who showed that a $(7/4-\varepsilon)$-approximation for sparse undirected unweighted Diameter requires $m^{4/3-o(1)}$ time under SETH. This corresponds to the $k=4$ point of the tradeoff and precedes the general construction studied in section 5.

For directed graphs, Li [Li21] and, independently, Dalirrooyfard and Wein [DW21] developed the full sequence of lower bounds: for every integer $k \geq 2$, a $(2-1/k-\varepsilon)$-approximation requires $m^{1+1/(k-1)-o(1)}$ time under SETH. The major remaining question was whether the same phenomenon could be established without using edge directions.

Dalirrooyfard, Li, and Vassilevska Williams [**DLW21**] answered this question affirmatively. Their result proves the same tradeoff for unweighted undirected graphs and simultaneously generalizes the previously known $k=2,3,4$ points. In particular, it condition-

ally establishes the optimality of the elementary factor-2 approximation in near-linear time.

Related variants. The same fine-grained viewpoint has been applied to several closely related distance problems. Abboud, Vassilevska Williams, and Wang [AVW16] studied Diameter and Radius in directed graphs as well as parameterized versions, giving both approximation algorithms and conditional lower bounds. Dalirrooyfard et al. [Dal+19] developed corresponding upper and lower bounds for ST-Diameter and bichromatic Diameter, and Berger, Kaufmann, and Vassilevska Williams [BKV23] pushed this line further with tight approximation algorithms for min-diameter and bichromatic min-diameter, the directed-graph analogue in which a pair's distance is taken as the smaller of its two directed distances.

Directed Diameter continues to exhibit a somewhat different algorithmic landscape. Abboud, Dalirrooyfard, Li, and Vassilevska Williams [Abb+23] obtained new subquadratic approximation algorithms and studied roundtrip Diameter. Their hardness results also use assumptions beyond SETH, including the All-Nodes k-Cycle hypothesis and hardness of approximate ℓ_∞-Closest-Pair. A further symptom of this different landscape is that the classical definition of directed diameter is simply infinite on graphs that are not strongly connected, which is the generic case for DAGs. Bals, Blikstad, Dadush, Nazari, and Schmidt [Bal+26] address this gap directly: they initiate the fine-grained study of the *reachability diameter*, the largest distance between a pair u, v with v reachable from u, giving both conditional lower bounds and additive approximations for this notion. These variants are outside the scope of this report, but they illustrate that Diameter, in its many guises, remains a central and still-active test problem for fine-grained complexity.

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